

Elijah E. Mullens
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EDUCATION

August 2022 -	Cornell University Ph.D., Astronomy and Space Sciences Thesis Title <i>Cloudy with a chance of Silicates: An Exploration of Exoplanet Aerosols and Weather</i> Thesis Advisor Dr. Nikole Lewis (nkl@cornell.edu) Master's Degree Received November 2024
August 2017 – May 2021	University of Florida <i>Graduated Cum Laude, Cumulative GPA 3.88</i> Bachelor of Science, Mathematics <i>Graduated Cum Laude, Major GPA 3.914</i> Bachelor of Science, Physics <i>Graduated Cum Laude, Major GPA 3.993</i> Bachelor of Science, Astrophysics <i>Graduated Cum Laude, Major GPA 3.993</i> Minor, East Asian Languages and Literatures – Japanese <i>Minor GPA: 3.596</i>
October 2019 – August 2020	Nagoya University (NUPACE Program) Certificate of Completion <i>Exchange Program, Concentration in Physics and Japanese</i>
2017	Tallahassee Community College / Florida State University Associate of Arts <i>Dual-Enrollment</i>

SELECTED PROPOSALS AND PUBLICATIONS

PI Proposals:

JWST Cycle 4 GO (ID 7686)

Brown Dwarf Broiler: Probing Chemical Quenching and Heat Redistribution in a Highly-Eccentric Brown Dwarf (PI: Elijah Mullens)

JWST Cycle 4 GO (ID 8309)

It's (poly)Morphin' Time! Solving the quartz quandary of WASP-17b (PI: Elijah Mullens)

[Cornell Chronicle Article on JWST Cycle 4 PI-ed Proposals](#)

Publications:

Mullens et al 2023

Characterizing the 3D Structure of Molecular Cloud Envelopes in the Cloud Factory Simulations 'ApJ 966:127

Mullens et al 2024

Implementation of Aerosol Mie Scattering in POSEIDON with Application to the Hot Jupiter HD 189733 b's Transmission, Emission, and Reflected Light Spectrum 'ApJ, 977:105

Mullens et al 2025

Life after Death: Europa in the evolving habitable zone of a Red Sun MNRAS, 540:1

[Space Article](#)

[Universe Today Article](#)

Mullens & Lewis 2025

Silicate Sundogs: Probing the Effects of Grain Directionality in Exoplanet Observations ApJL, 988:L43

[Cornell Chronicle Article](#)

Co-authored Publications w/ Significant Contribution:

Grant et al 2023

JWST-TST DREAMS: Quartz Clouds in the Atmosphere of WASP-17b 'ApJL, 956:L32

[Press Release](#)

Valentine et al 2024

JWST-TST DREAMS: Nonuniform Dayside Emission for WASP-17b from MIRI/LRS 'AJ 168:123

Louie et al 2025

JWST-TST DREAMS: A Precise Water Abundance for Hot Jupiter WASP-17b from the NIRISS SOSS Transmission Spectrum 'AJ, 169:86

CONFERENCES

April 2023 Protostars and Planets 7 (Poster presentation)

'Unveiling the Nature of Diffuse Interstellar Envelopes Around Dense Star-forming Clouds'

January 2023 AAS 241 (iPoster presentation)

'Unveiling the Nature of Diffuse Interstellar Envelopes Around Dense Star-forming Clouds'

January 2024 AAS 243 (iPoster presentation)

'A New Atmospheric Retrieval Paradigm for Exoplanet Aerosols'

June 2024 AbGradCon 2024 (Organizer + Poster presentation)

'Life After Death: Europa in the Habitable Zone of a Red Sun and White Dwarf'

June 2024 OPAG (Lighting Talk + Poster presentation)

'Life After Death: Europa in the Habitable Zone of a Red Sun and White Dwarf'

July 2024 ERES (Accepted Talk)

'A New Atmospheric Retrieval Paradigm for Exoplanet Aerosols'

November 2024 Unearthing, Unseeing: Archaeology (Attendee)

January 2025 MLA 2025 (Attendee)

June 2025	AAS 246	(Lab Astro Workshop Lighting Talk)
'The Quartz Quandary of WASP-17b'		
June 2025	AAS 246	(iPoster presentation)
'Silicate Sundogs and Sandy Thunderstorms: Probing the Effects of Grain Directionality and Chemistry in Exoplanet Observations'		
July 2025	OWL 2025	(Attendee w/ full funding)
December 2025	AGU 2025	(Accepted Talk)
'HEL-Fire and Brimstone: High Temperature Optical Properties of Quartz and Tridymite for Substellar Cloud Studies'		
January 2026	MLA 2026	(Accepted Talk)
'An idea you'll have to give birth to': An exploration of astronomy and alternative relationality in Planet-Size X-Men		

HONORS AND AWARDS

Honors: Dean's List, President's Honor Roll (*University of Florida*)

Scholarships:

2017 – 2021	Presidential Scholarship, <i>University of Florida</i>
2017 – 2021	Florida Bright Futures Scholarship Academic Scholars Award
2019 – 2020	JASSO Scholarship, <i>Nagoya University</i>

Fellowships:

2022 – 2023	Deans Excellence Fellowship, <i>Cornell University</i>
2023 –	NSF GRFP, Grant No. 2139899

Grants:

2023	Cornell University Conference Grant (for PP7)
2024	Cornell University Conference Grant (for AAS243)
2025	Cornell University Research Travel Grant (for NASA internship)

Awards:

2025	Cranson W. and Edna B. Shelly Award – Outstanding Teaching Award
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CO-AUTHORED PUBLICATIONS

Gressier et al 2025
JWST-TST DREAMS: A Supersolar Metallicity in WASP-17 b's Dayside Atmosphere from NIRISS SOSS Eclipse Spectroscopy AJ 169:57

Espinoza et al 2025
JWST-TST DREAMS: NIRSpec/PRISM Transmission Spectroscopy of the Habitable Zone Planet TRAPPIST-1 e ApJL 990:L52

Glidden et al 2025

JWST-TST DREAMS: Secondary Atmosphere Constraints for the Habitable Zone Planet TRAPPIST-1 e ApJL 990:L53

Gressier et al 2025

JWST TST-DREAMS: Sulfur Dioxide in the Atmosphere of the Neptune-mass Planet HAT-P-26b from NIRSpec G395H Transmission Spectroscopy AJ 170:292

Lustig-Yaeger et al 2025

JWST-TST DREAMS: The Nightside Emission and Chemistry of WASP-17b ApJ 994:L4

Meynardie et al 2025

Ross 458 C: Gas Giant or Brown Dwarf? ApJ 994:237

Sotzen et al 2025

Non-detection of Nightside Emission from HAT-P-26b due to Stellar Contamination RNAAS 9:352

WORK AND RELEVANT EXPERIENCE

June 2025 –
August 2025

NASA Goddard Space Flight Center

Intern (PI: Sarah Moran)

Experimental and Modeling Investigation of Exoplanet Cloud Properties

August 2024 –
May 2025

Cornell University Astronomy Department

Teaching Assistant

- ASTRO 1195 – Observational Astronomy
 - Smaller class (30 students). Held night labs, graded homework, and taught a few guest lectures.
- ASTRO 1102 (Head – TA) – Our Solar System
 - Large class (130 students). Was in charge of TAs with multiple duties such as: attending lecture, holding office hours, developing material for discussion section, organizing TA duties.

June 2022 –
August 2022

Space Telescope Science Institute

Space Astronomy Summer Program Intern

- Worked on a project titled ‘Unveiling the Nature of Diffuse Interstellar Envelopes Around Dense Star-forming Clouds’ (PI: Catherine Zucker)
- Developed a pipeline to compare interstellar cloud simulations to observations, gave a presentation at the 2022 SASP Symposium, will give a poster presentation at AAS 241 and Protostars and Planets VII. Resulting paper accepted for publication in ApJ.

December 2021 –
June 2022

Challenger Learning Center of Tallahassee

Planetarium Instructor

- Created planetarium shows using Digistar 3, runs planetarium shows, and prepares and presents the monthly ‘Monthly Skies Over Tallahassee’ and

other educational pre-shows ('Monthly Astronomy Learning Topic', 'Space News', 'Monthly Solar System Object') live

2019 – 2020

English Camp Tutor

- Organized by Meitetsu World Travels Inc. Conducted at Nagoya University
- Led and organized group activities for over 1,000 children
- Gave presentation on home country and taught basic math

October 2021 –
June 2022

Volunteer Research Assistant

Nagoya University, Department of Physics

- Assistant to Professor Hiroshi Kobayashi on the origin of planetary systems. Achieved an understanding on creating orbital simulations; creating a comprehensive model for planet formation and Galilean moon formation; and modelling post-main sequence planetary system evolution (both compositional and orbital) and second-generation planetary formation around white dwarfs.

August 2020 –
December 2020

Undergraduate Researcher

University of Florida, Department of Astronomy

- Contributed to Professor Charles Telesco's IMPS (Integrated Miniature Polarimeter and Spectrograph) project. Achieved an understanding of the instrument while learning further topics on polarimetry and astrobiology.

July 2019 –
February 2020

Individual Work

Nagoya University, Department of Physics

- Further topics in Statistical Physics with Professor John Wojdylo, covering microcanonical and canonical formalism, quantum fluids, and mean field theory.

Nagoya University, Department of Earth and Environmental Sciences

- Further topics in Earth and Planetary Sciences with Professor Marc Humblet, research paper written on TESS and planetary detection methods.

University of Florida, Department of Mathematics

- Differential Geometry with Professor Luca Di Cerbo covering Do Carmo's *Differential Geometry of Curves and Surfaces*, with a presentation at the end.

CERTIFICATES

December 2021– April 2021	Florida State University Global Partners Certificate Equity, Diversity, and Inclusion Certificate
August 2021 – October 2021	Coursera The University of Sydney: Data-driven Astronomy
October 2019 – August 2020	NUPACE Program Nagoya University: Certificate of Completion

LANGUAGES

- English (*Native*)
- Japanese (*JLPT N3 level, lived in Japan for six months, 5 semesters taken*)

ORGANIZATIONAL MEMBERSHIP/VOLUNTEER WORK

August 2024 – August 2025	Graduate Mentor To: Isabella Huckabee
September 2023 – May 2024	GRASHOPR Graduate Student School Outreach Program.
August 2024 –	Undergraduate Mentor To: Ian Branigan
June 2023 –	Cornell University Astronomy Department Two outreach events: middle school field trip and annual 4-H event
February 2022 – August 2022	FSU Astronomy Club Helped found the club and organize events at the FSU observatory for both high schoolers and college students. Helped set up the observatory for undergraduate astronomy labs.
February 2022 – August 2022	Sea-to-See Program Marine biology outreach program. Helped collect and take care of marine specimens from beaches in hometown. Helped set-up and run booths at local STEAM expos with marine specimen showcases.
November 2021 – August 2022	Challenger Learning Center of Tallahassee Volunteered at the James Webb Space Telescope (JWST) Pre-Launch Party.
September 2021 – August 2022	Tallahassee Astronomical Society Helped set up and facilitate demonstrations using sun-filler telescopes at the JWST Pre-Launch Party.

August 2021 – August 2022	Astronomers for Planet Earth
August 2021 – August 2022	Tallahassee Scientific Society Ran and organized a booth with multiple physical science demonstrations (aimed at 3 rd graders) for the 10 th Annual Tallahassee Science Fair. Helped organize local science fairs. Volunteered as a science fair judge at the Tallahassee School of Math and Science’s science fair, and the Capital Regional Science and Engineering Fair.
August 2021 – August 2022	oSTEM <i>oSTEM Global Chapter</i>
August 2019 – February 2020	名大セクマイサークル『雨玉』 Circle at Nagoya University “Ametama” <i>Nagoya University</i>
August 2019 – February 2020	Language Exchange Club <i>Nagoya University</i>
August 2019 – February 2020	留学サークル Foreign Students Club <i>Nagoya University</i>
August 2018 – May 2019	oSTEM Non-profit professional association <i>University of Florida</i>
August 2017 – May 2018	J-Club Japanese Language and Culture Club <i>University of Florida</i>
August 2017 – May 2018	Model UN GatorMUN 2018 <i>University of Florida</i>

REFERENCES

Nikole Lewis

Associate Professor

Cornell University, Department of Astronomy

Nikole.lewis@cornell.edu

Catherine Zucker

Astrophysicist

Smithsonian Astrophysical Observatory

Catherine.Zucker@cfa.harvard.edu

John Andrew Wojdylo

Designated Professor (G30)

Nagoya University, Department of Physics

Solid State Theory Laboratory (Condensed-Matter Theory Group)

John.wojdylo@s.phy.nagoya-u.ac.jp

Bonnie Halsell

Office Manager

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